StudyBuddy

AI Study Coach Agent

An AI-agent web application built with Node.js, Express and the Google Gemini API

Live application

https://studybuddy-XXXX.onrender.com

Source code repository

https://github.com/manganjokean/studybuddy

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Abstract

StudyBuddy is a publicly deployed web application that helps students turn their own lecture notes into interactive study material. The project implements an *AI agent* rather than a plain chatbot: a Node.js/Express back end exposes a set of tools to the Google Gemini model, and an agentic loop lets the model decide which tool to call, execute it on the server, feed the result back, and only then produce a final answer for the user. The client side is a chat interface written in vanilla HTML, CSS and JavaScript: user and agent turns appear as bubbles, the agent's tool calls are animated as a visible plan, and the quiz, flashcards or explanation built by a tool are rendered inline in the same conversation. Quiz results are persisted in the browser with `localStorage`. The application is deployed on Render and the Gemini API key is kept entirely outside the repository through environment variables, a strict `.gitignore` and an automated pre-push secret scanner. This report describes the requirements, architecture, agent design, implementation, deployment, security measures, testing, and lessons learned during the project.

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1 Introduction

Students routinely revise from passive material: re-reading slides, highlighting text and re-reading again. Research on learning consistently favours *active recall* — retrieving information from memory — but producing practice questions from one's own notes is tedious.

StudyBuddy addresses this with a small, focused web application. A student pastes their notes (or types a topic), and an AI agent decides how best to help: it can generate a quiz, build a flashcard deck, or explain a concept at an appropriate level. The distinctive feature of the project is that the model does not simply emit text: it *calls tools* that the server executes, and the structured output of those tools drives the interactive front end.

Project objectives

- Build a working web application whose front end demonstrates client-side skills (DOM manipulation, `fetch`, event handling, state, local persistence, responsive CSS).
- Implement an AI agent with a real tool-calling loop, not a single chat completion.
- Deploy the application to a public host so it can be opened by anyone.
- Keep the API key out of source control and out of the browser.
- Produce this report, linked to the live application and the source repository.
- Present and demonstrate the application in class.

2 Requirements

Table 1: Functional and non-functional requirements.

ID	Requirement	Status
FR-1	Accept study material or a free-text topic from the user	Implemented
FR-2	Agent selects and executes at least two distinct tools	Implemented (3 tools)
FR-3	Display the agent's plan / tool-call trace to the user	Implemented
FR-4	Interactive quiz with per-question feedback and a final score	Implemented
FR-5	Flashcard deck with flip interaction	Implemented
FR-6	Structured concept explanation	Implemented
FR-7	Persist quiz history locally in the browser	Implemented (`localStorage`)
FR-8	Graceful error handling for invalid input, rate limits and API failures	Implemented
NFR-1	Deployed to a public URL	Render
NFR-2	No API key in the repository or in client-side code	Enforced by tooling
NFR-3	Responsive layout, no front-end framework	Vanilla HTML/CSS/JS

3 Technology Stack

Table 2: Technologies used.

Layer	Technology	Reason
Front end	HTML5, CSS3, vanilla ES6 JS	Course focus: client-side programming
Back end	Node.js 22+ / Express 5	Simple static + JSON API server
AI model	Google Gemini (`gemini-3.8-flash`)	Native function calling, free tier
API style	REST (`generateContent`)	No SDK dependency, transparent requests
Hosting	Render	Free tier, Node runtime, environment variables
Versioning	Git / GitHub	Required repository link
Report	LaTeX on Overleaf	PDF output (required deliverable)

4 System Architecture

The application is a classic two-tier structure. The browser never talks to the AI provider: all model calls go through the Express back end, which is the only component that holds the API key.

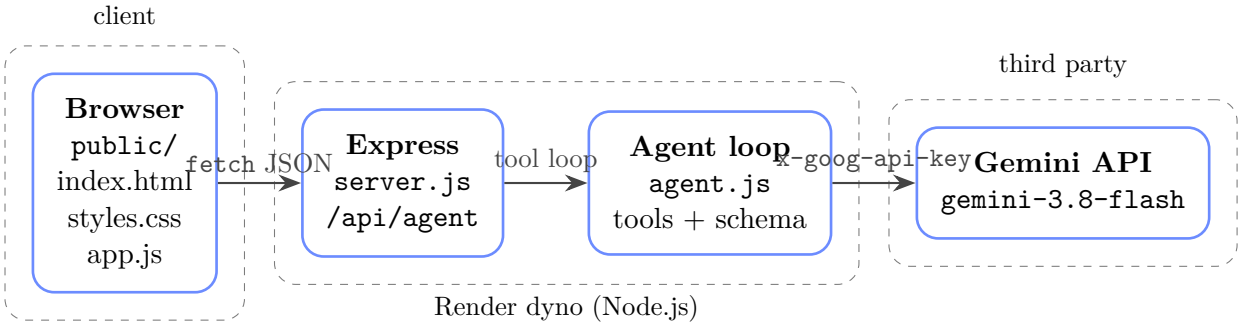


Figure 1: Request flow. The API key exists only inside the Render environment.

Project structure

```
studybuddy/
|-- server.js            # Express server, static hosting, /api/agent
|-- agent.js             # agent loop, tool schemas, tool implementations
|-- scripts/keycheck.js  # pre-push secret scanner
|-- public/
|   |-- index.html       # markup: input, trace, tabs, quiz, cards
|   |-- styles.css       # theming, quiz UI, 3D flip cards, responsive
|   `-- app.js           # fetch, trace animation, quiz logic, storage
|-- report/              # this LaTeX report + figures
|-- .env                 # API key (gitignored, never pushed)
`-- .env.example         # placeholder committed on purpose
```

5 AI Agent Design

What makes it an agent

A chatbot performs a single request/response and returns prose. StudyBuddy's back end instead runs a *loop*:

1. The user goal, conversation history and a system prompt are sent to Gemini together with a list of **function declarations** (the tool schemas).
2. If the model answers with a `functionCall`, the server executes that tool locally, appends a `functionResponse` containing the real result, and calls the model again.
3. Steps repeat until the model replies with plain text, or a safety limit of four rounds is reached.
4. The response returned to the browser contains three parts: the agent's natural-language reply, the **trace** of tools invoked, and the raw structured **results** used to render the interactive components.

Listing 1: Simplified agent loop (agent.js).

```
for (let step = 0; step < MAX_STEPS; step++) {
  const parts = await callGemini({ contents, tools: TOOLS,
                                   systemInstruction: SYSTEM_PROMPT });
  const calls = parts.filter(p => p.functionCall);
  if (!calls.length) { finalText = joinText(parts); break; }  // done

  contents.push({ role: 'model', parts });
  for (const c of calls) {
    const output = await TOOL_IMPLS[c.functionCall.name](c.functionCall.args);
    results[c.functionCall.name] = output;                    // structured result
    contents.push({ role: 'user', parts: [{ functionResponse: ... }] });
  }
}
```

Available tools

Table 3: Tools exposed to the model.

Tool	Arguments	Behaviour
`generateQuiz`	`source, topic, count, difficulty`	Calls Gemini with a JSON response schema; validates each question and clamps `count` to 3–10
`generateFlashcards`	`source, topic, count`	Produces front/back card pairs, filtered for well-formed entries
`explainConcept`	`concept, level`	Free-text answer forced into `##` sections so the client can render it reliably

Robustness measures

- **Structured output**: quizzes and flashcards are requested as JSON with an explicit schema, and every returned field is validated before it reaches the browser, instead of trusting the model to format things correctly.
- **Input clamping**: question counts, string lengths and history size are bounded before they reach the API.
- **Graceful degradation**: if the final summarising call fails (for example due to a rate limit), the already-produced structured results are still returned with a fallback message, so the user is not left with an error screen.
- **Friendly errors**: HTTP 429, 400/403 and network failures are mapped to plain-language messages; technical details are written to the server log only.

6 Client-Side Implementation

The graded focus of the course is client-side development, so the front end is written without any framework.

- **Conversation thread**: user and agent turns are rendered as chat bubbles with a typing indicator while the agent works, and the thread auto-scrolls to the newest message.
- **Agent trace animation**: each tool call returned by the API is appended to an "Agent plan" block one step at a time, so the audience can watch the agent decide and act before the answer arrives.
- **Inline artefact cards**: the quiz, flashcard deck or explanation produced by a tool is injected directly into the same conversation turn – no page change, no separate results screen.
- **Quiz engine**: one question at a time, immediate green/red feedback together with the explanation, a progress pill, and a conic-gradient score ring at the end.
- **Flashcards**: a 3D `rotateY` flip implemented with `transform-style: preserve-3d` and `backface-visibility`, plus a Fisher–Yates shuffle.
- **State & persistence**: scores are written to `localStorage` under `studybuddy.scores`; the last eight conversation turns are sent with every request so the agent can answer follow-ups in context.
- **Composer**: *Enter* sends, *Shift+Enter* inserts a newline, the textarea grows with its content, and mode/level/item selectors steer the agent's tool choice.
- **Resilience**: spinner on the send button, toast notifications for errors, and all dynamic text inserted through an HTML-escaping helper to avoid injection from model output.
- **Responsive layout**: a single-column chat shell that adapts below 780 px, with touch-friendly card grids.

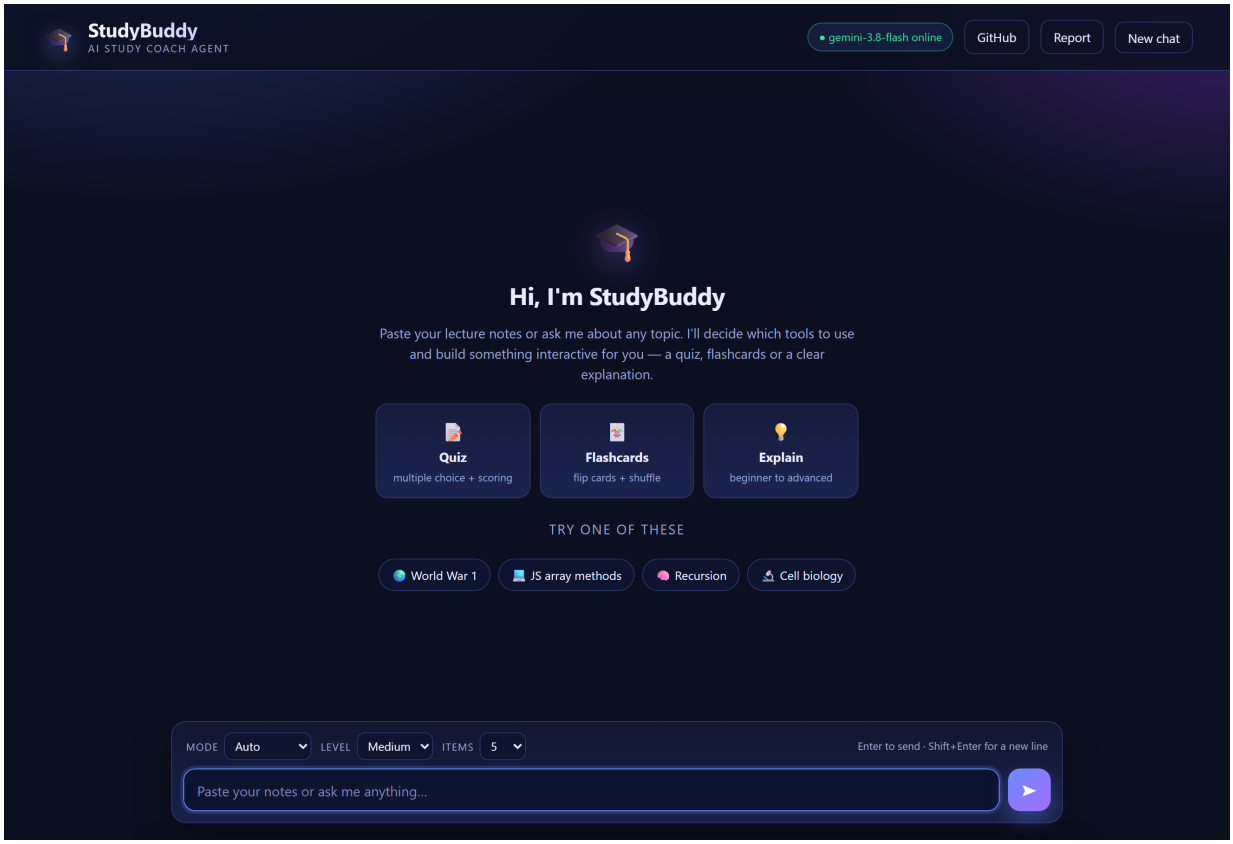


Figure 2: Landing screen: capability overview, example prompts and the composer.

7 Deployment

The application is deployed on **Render** as a Node web service. Heroku, also listed in the brief, removed its free tier, while Render provides a free plan that runs a long-lived Express process and injects environment variables.

1. Push the repository to GitHub: `https://github.com/manganjokean/studybuddy`.
2. In Render: *New* → *Web Service*, connect the repository.
3. Build command `npm install`, start command `npm start`.
4. Under *Environment*, add `GEMINI_API_KEY` with the secret value. This is the *only* place the key is stored outside the developer's machine.
5. Deploy. The service is reachable at `https://studybuddy-XXXX.onrender.com`.

Free-tier caveat. The Render free plan puts dynos to sleep after inactivity, so the first request after idle time takes roughly 20–30 seconds. For a live presentation the application is opened (and kept in a background tab) a few minutes before the demo begins.

8 Security: API Key Handling

The brief explicitly forbids pushing API keys to GitHub. This was treated as a design constraint rather than an afterthought, and enforced at four levels:

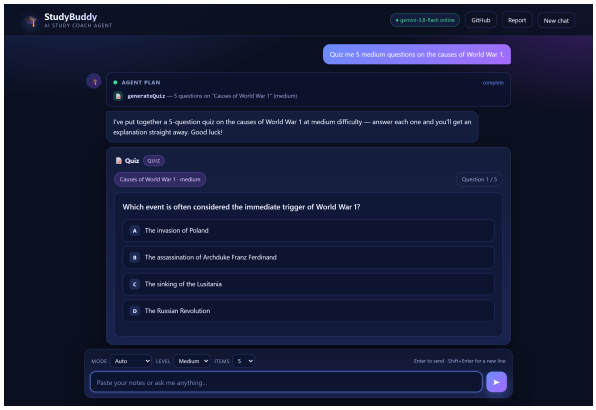


Figure 3: Chat turn: user request, agent plan and reply.

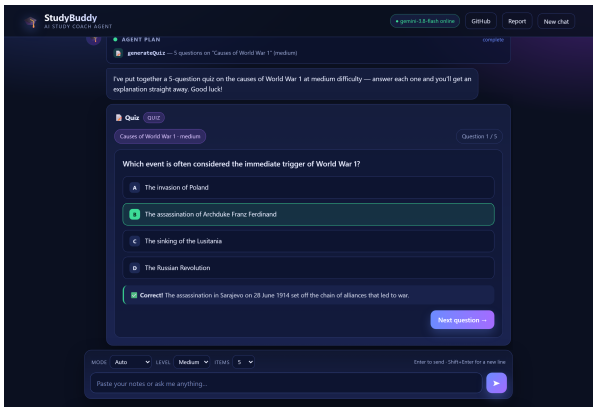


Figure 4: Inline quiz with per-question feedback.

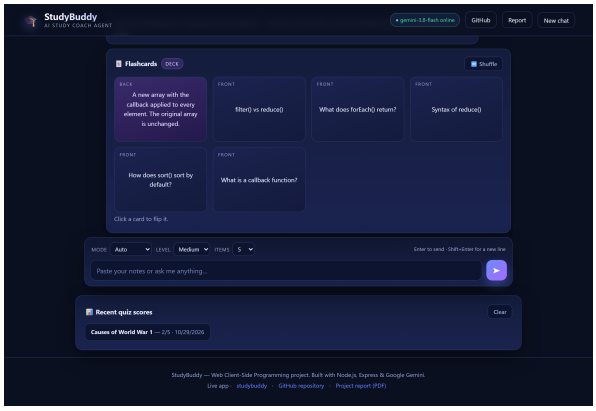


Figure 5: Flashcard deck with flip interaction.

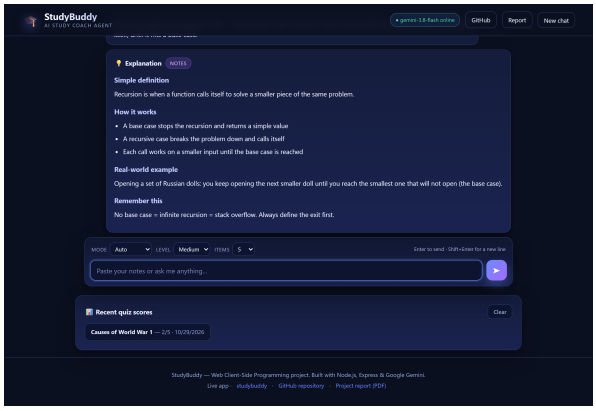


Figure 6: Structured concept explanation.

Listing 2: Verifying the repository before every push.

```
$ npm run keycheck
> node scripts/keycheck.js
  No API keys or secrets found in the project. Safe to push.
```

Additionally, `.env.example` is committed with placeholder values only, which documents the required configuration without disclosing anything. The server also detects when the placeholder was never replaced and reports *"API key missing"* in the health endpoint instead of failing obscurely at request time.

9 Testing

10 Challenges and Lessons Learned

- **Express 5 breaking change.** The legacy wildcard route `app.get('*')` throws `Missing parameter name` under Express 5's `path-to-regexp` parser. The SPA fallback was rewritten as a terminal `app.use()` handler.
- **Structured output is not optional.** Asking the model for free-form JSON failed intermittently; requesting `responseMimeType` plus an explicit schema, and then validating every field, removed those failures almost entirely.

Table 4: Defence-in-depth for the secret.

Layer	Mechanism
Repository	`.gitignore` written *before* the first commit; it lists `.env`, `.env.local`, `.env.*.local` and `node_modules/`
Code	The key is read only via `process.env.GEMINI_API_KEY`; there is no fallback literal anywhere in the source
Transport	The key is sent as the `x-goog-api-key` header directly to Google's API and is never returned in any HTTP response
Tooling	`npm run keycheck` scans every text file for credential patterns (Google `AIza`, OpenAI/Anthropic `sk-`, bearer tokens, assigned key variables) and exits non-zero if anything matches

- **Counting API calls.** The naive loop cost three model round-trips per request, which is slow on a free tier. Structured results are therefore returned to the client directly from the tool, and only the short conversational summary goes back through the model.
- **Secrets discipline.** Writing `.gitignore` first and adding an automated scanner turned "don't leak the key" from a reminder into an enforced rule.
- **Free-tier realities.** Cold starts and rate limits must be planned for when the application is demonstrated live.

11 Conclusion

StudyBuddy meets every requirement of the brief: it is a working web application built with a JavaScript framework-free client and an Express back end, it uses an AI agent with a genuine tool-calling loop, it is deployed publicly, it keeps its API key out of GitHub by construction, and this report links to both the live application and the repository.

The most valuable outcome was understanding the difference between a chatbot and an agent: giving the model *actions* to perform, and validating what comes back, is what allowed a single conversation to produce interactive artefacts the browser could render. Future work would add spaced-repetition scheduling for flashcards, server-side persistence of study history, and streaming responses so the plan trace appears in real time.

Links

- Live application: `https://studybuddy-XXXX.onrender.com`
- Source repository: `https://github.com/manganjokean/studybuddy`
- This report (PDF): `report/main.pdf`

Table 5: Test cases executed before submission.

Area	Case	Result
Health	`GET /api/health` reports model and key status	Pass
Validation	Empty message returns `400` with a clear error	Pass
Validation	Oversized message rejected (`413`)	Pass
Errors	Malformed JSON body returns `400`, not `500`	Pass
Errors	Missing key returns an actionable message	Pass
Errors	Rate limit (HTTP 429) mapped to friendly text	Pass
Routing	Unknown `/api` route returns JSON `404`	Pass
Routing	Deep links fall back to the single-page app	Pass
Agent	Quiz generated, validated and rendered interactively	Pass
Agent	Flashcards flip, shuffle and re-render	Pass
Agent	Score written to and read from `localStorage`	Pass
Security	`git ls-files` contains no `.env`	Pass

References

[1] Google AI for Developers, *Generating content — Gemini API*, https://ai.google.dev/api/generate-content, accessed 2026.

[2] ExpressJS, *Express 5.x Guide*, https://expressjs.com/, accessed 2026.

[3] Render, *Documentation — Web Services*, https://render.com/docs/web-services, accessed 2026.

[4] S. Yao et al., *ReAct: Synergizing Reasoning and Acting in Language Models*, ICLR 2023.

[5] N. C. Dingemanse, "Retrieval practice is more effective than repeated study," *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 2020.